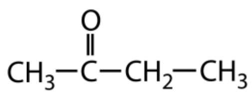
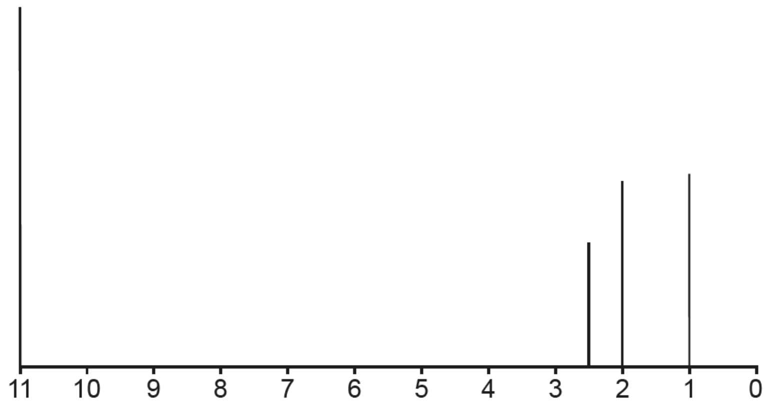
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)			no reaction with dichromate → not 1° or 2° alcohol or aldehyde /		1				
				could be 3° alcohol or ketone or carboxylic acid(1)		1				2
				MS molecular ion peak → M_r 72 (1)	1					
				MS fragments 15 → CH_3^+ 29 → $C_2H_5^+$ 43 → $CH_3C{=}O^+$ 57 → $C_2H_5C{=}O^+$ (any two) (1)		1				
				IR has no OH peak → not acid (1)			1			
				IR peak at 1750 → C=O present(1)			1			
				^{13}C NMR 4 peaks → four carbon environments (1) peak at 30/40 → must be ketone OR peak at 210 → must be aldehyde or ketone (1)			1			
				molecular formula C_4H_8O (1)			1	8		
				award any seven of eight possible marks						
				final mark reserved for correct compound						
				X is  (1) accept butanone						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			 total three peaks (1) (from left to right ⇨ 2.0 to 3.0, 2.0 to 2.5 and 0.1 to 2.0) height ratio 2 : 3 : 3 (1) ecf possible if incorrect compound given in part (a) e.g. butanal → four peaks, ratio 3:2:2:1			2	2		
				Question 11 total	**1**	**3**	**6**	**10**	**0**	**2**